API Documentation

Banking Loan Risk Analysis Application | FastAPI REST API

Base URL: http://127.0.0.1:8000 **Interactive OpenAPI:** /docs **OpenAPI JSON:** /openapi.json

This document covers routes defined in the uploaded source repository. Example values below are illustrative, not captured live API responses. Protected endpoints require **Authorization: Bearer <access_token>**.

1. API Endpoint Inventory

Method	Endpoint	Purpose
GET	/	Root
GET	/health	Health Check
GET	/health/db	Database Health Check
POST	/api/auth/login	Login
GET	/api/auth/me	Get My Profile
GET	/api/dashboard/portfolio	Get Portfolio Summary
GET	/api/dashboard/risk-distribution	Get Risk Distribution
GET	/api/dashboard/monthly-financials	Get Monthly Financials
GET	/api/dashboard/segment-risk	Get Segment Risk
GET	/api/customers	Get Customers
GET	/api/customers/{customer_id}	Get Customer Details
POST	/api/reviews	Create Review
GET	/api/reviews	Get Reviews
GET	/api/reviews/{review_id}	Get Review
PATCH	/api/reviews/{review_id}	Update Review
DELETE	/api/reviews/{review_id}	Delete Review
GET	/api/exports/customers	Export Customers

2. Authentication and Request Conventions

Use **POST /api/auth/login** with application/x-www-form-urlencoded fields **username** and **password**. Supported configured usernames are analyst and administrator. The token returned is passed to protected endpoints. Login does not accept a JSON body.

GET requests use query parameters. POST and PATCH review endpoints use JSON. CSV export returns text/csv. Numeric financial values are represented by JSON numbers or decimal-compatible values according to response serialization.

3. Detailed API Reference

3.1 GET /

Purpose: root. **Source:** main.py.

Request format: Path/query parameters; no body.

Response format: JSON response.

Example request

```
GET / HTTP/1.1
```

Illustrative response

```
{"message":"Banking Loan Risk Analysis API","docs":"/docs"}
```

3.2 GET /health

Purpose: health check. **Source:** main.py.

Request format: Path/query parameters; no body.

Response format: HealthResponse.

Example request

```
GET /health HTTP/1.1
```

Illustrative response

```
{"status":"healthy","message":"Banking Loan Risk Analysis API is running"}
```

3.3 GET /health/db

Purpose: database health check. **Source:** main.py.

Request format: Path/query parameters; no body.

Response format: DatabaseHealthResponse.

Example request

```
GET /health/db HTTP/1.1
```

Illustrative response

```
{"status":"healthy","database":"bank_risk","message":"MySQL database connection successful"}
```

3.4 POST /api/auth/login

Purpose: login. **Source:** routers/auth.py.

Request format: application/x-www-form-urlencoded (username, password).

Response format: TokenResponse.

Example request

```
POST /api/auth/login
Content-Type: application/x-www-form-urlencoded

username=analyst&password=<password>
```

Illustrative response

```
{"access_token":"<jwt>","token_type":"bearer","expires_in":1800,"role":"analyst"}
```

3.5 GET /api/auth/me

Purpose: get my profile. **Source:** routers/auth.py.

Request format: Path/query parameters; no body.

Response format: AuthenticatedUser.

Example request

```
GET /api/auth/me
Authorization: Bearer <jwt>
```

Illustrative response


```
{"username":"analyst","role":"analyst"}
```


3.6 GET /api/dashboard/portfolio

Purpose: get portfolio summary. **Source:** routers/dashboard.py.

Request format: Path/query parameters; no body.

Response format: PortfolioSummaryResponse.

Example request

```
GET /api/dashboard/portfolio
Authorization: Bearer <jwt>
```

Illustrative response


```
{"summary_id":1,"total_customers":30000,"total_credit_exposure":5000000000,
"average_credit_limit":166666.67,"average_risk_score":45.2,"total_defaulted_customers":6636,
"historical_default_rate":0.2212,"last_refreshed_at":"2026-10-11T10:00:00"}
```


3.7 GET /api/dashboard/risk-distribution

Purpose: get risk distribution. **Source:** routers/dashboard.py.

Request format: Path/query parameters; no body.

Response format: list[RiskDistributionResponse].

Example request

```
GET /api/dashboard/risk-distribution
Authorization: Bearer <jwt>
```

Illustrative response


```
[{"risk_band":"HIGH","customer_count":5000,"percentage":16.67,"average_risk_score":78,
"total_credit_exposure":850000000,"defaulted_customers":1800,"historical_default_rate":0.36,
"last_refreshed_at":"2026-10-11T10:00:00"}]
```


3.8 GET /api/dashboard/monthly-financials

Purpose: get monthly financials. **Source:** routers/dashboard.py.

Request format: Path/query parameters; no body.

Parameters: month (str), **sort_by** (Literal['month', 'customer_count), **sort_order** (Literal['asc', 'desc'])

Response format: list[MonthlyFinancialResponse].

Example request

```
GET /api/dashboard/monthly-financials?sort_by=month&sort_order=asc
Authorization: Bearer <jwt>
```

Illustrative response


```
[{"month":"2025-09","customer_count":30000,"total_bill_amount":10000000,
"total_payment_amount":8000000,"average_bill_amount":333.33,"average_payment_amount":266.67,
"payment_to_bill_ratio":0.8,"last_refreshed_at":"2026-10-11T10:00:00"}]
```


3.9 GET /api/dashboard/segment-risk

Purpose: get segment risk. **Source:** routers/dashboard.py.

Request format: Path/query parameters; no body.

Parameters: segment_type (str), **segment_value** (str), **sort_by** (Literal['segment_type', 'segment), **sort_order** (Literal['asc', 'desc'])

Response format: list[SegmentRiskResponse].

Example request

```
GET /api/dashboard/segment-risk?segment_type=education&sort_order=desc
Authorization: Bearer <jwt>
```

Illustrative response

```
[{"segment_type":"education","segment_value":"University","customer_count":1000,
"high_risk_customers":220,"high_risk_percentage":22,"average_risk_score":52,
"average_delayed_months":1.5,"historical_default_rate":0.2,"total_credit_exposure":160000000,
"last_refreshed_at":"2026-10-11T10:00:00"}]
```

3.10 GET /api/customers

Purpose: get customers. **Source:** routers/customers.py.

Request format: Path/query parameters; no body.

Parameters: page (int), **page_size** (int), **risk_band** (Literal['LOW', 'MEDIUM', 'HIGH']), **gender_label** (str), **education_label** (str), **marriage_label** (str), **age_group** (str), **min_age** (int), **max_age** (int), **min_credit_limit** (Decimal), **max_credit_limit** (Decimal), **sort_by** (Literal['customer_id', 'age', 'c), **sort_order** (Literal['asc', 'desc'])

Response format: PaginatedCustomerResponse.

Example request

```
GET /api/customers?page=1&page_size=2&risk_band=HIGH&sort_by=risk_score&sort_order=desc
Authorization: Bearer <jwt>
```

Illustrative response

```
{"total_records":1,"page":1,"page_size":2,"total_pages":1,"data":[{"customer_id":1,
"gender_label":"Female","education_label":"University","marriage_label":"Single","age":30,
"age_group":"30-39","credit_limit":50000,"months_with_delay":3,
"latest_credit_utilization":0.8,"payment_to_bill_ratio":0.25,"default_next_month":1,
"risk_score":85,"risk_band":"HIGH","risk_reason":"Repayment delays"}]}
```

3.11 GET /api/customers/{customer_id}

Purpose: get customer details. **Source:** routers/customers.py.

Request format: Path/query parameters; no body.

Parameters: customer_id (int)

Response format: CustomerDetailResponse.

Example request

```
GET /api/customers/1
Authorization: Bearer <jwt>
```

Illustrative response

```
{"customer":{"customer_id":1,"gender":2,"gender_label":"Female","education":2,
"education_label":"University","marriage":2,"marriage_label":"Single","age":30,
"age_group":"30-39"},"credit_profile":null,"risk_assessment":null,"bills":[],"payments":[]}
```

3.12 POST /api/reviews

Purpose: create review. **Source:** routers/reviews.py.

Request format: JSON body (ReviewCreate).

Parameters: review (ReviewCreate)

Response format: ReviewResponse.

Example request

```
POST /api/reviews
Authorization: Bearer <jwt>
Content-Type: application/json
```

```
{"customer_id":1,"review_status":"Pending","priority":"High","review_notes":"Manual
verification needed"}
```

Illustrative response

```
{"review_id":10,"customer_id":1,"review_status":"Pending","priority":"High",
"review_notes":"Manual verification needed","review_date":"2026-10-11T10:00:00",
"updated_at":"2026-10-11T10:00:00"}
```

3.13 GET /api/reviews

Purpose: get reviews. **Source:** routers/reviews.py.

Request format: Path/query parameters; no body.

Parameters: page (int), **page_size** (int), **customer_id** (int), **review_status** (Literal['Pending', 'In Progress'), **priority** (Literal['Low', 'Medium', 'High']), **sort_by** (Literal['review_id', 'customer_i), **sort_order** (Literal['asc', 'desc'])

Response format: PaginatedReviewResponse.

Example request

```
GET /api/reviews?page=1&page_size=20&review_status=Pending
Authorization: Bearer <jwt>
```

Illustrative response

```
{"total_records":1,"page":1,"page_size":20,"total_pages":1,"data":[{"review_id":10,
"customer_id":1,"review_status":"Pending","priority":"High","review_notes":"Manual
verification needed","review_date":"2026-10-11T10:00:00",
"updated_at":"2026-10-11T10:00:00"}]}
```

3.14 GET /api/reviews/{review_id}

Purpose: get review. **Source:** routers/reviews.py.

Request format: Path/query parameters; no body.

Parameters: review_id (int)

Response format: ReviewResponse.

Example request

```
GET /api/reviews/10
Authorization: Bearer <jwt>
```

Illustrative response

```
{"review_id":10,"customer_id":1,"review_status":"Pending","priority":"High",
"review_notes":"Manual verification needed","review_date":"2026-10-11T10:00:00",
"updated_at":"2026-10-11T10:00:00"}
```

3.15 PATCH /api/reviews/{review_id}

Purpose: update review. **Source:** routers/reviews.py.

Request format: JSON body (ReviewUpdate; optional fields).

Parameters: review (ReviewUpdate), **review_id** (int)

Response format: ReviewResponse.

Example request

```
PATCH /api/reviews/10
Authorization: Bearer <jwt>
Content-Type: application/json

{"review_status":"Completed","review_notes":"Reviewed"}
```

Illustrative response

```
{"review_id":10,"customer_id":1,"review_status":"Completed","priority":"High",
```

```
"review_notes":"Reviewed","review_date":"2026-10-11T10:00:00",
"updated_at":"2026-10-11T10:00:00"}
```

3.16 DELETE /api/reviews/{review_id}

Purpose: delete review. **Source:** routers/reviews.py.

Request format: Path/query parameters; no body.

Parameters: review_id (int)

Response format: 204 No Content.

Example request

```
DELETE /api/reviews/10
Authorization: Bearer <administrator-jwt>
```

Illustrative response

```
HTTP 204 No Content (empty body)
```

3.17 GET /api/exports/customers

Purpose: export customers. **Source:** routers/exports.py.

Request format: Path/query parameters; no body.

Parameters: risk_band (Literal['LOW', 'MEDIUM', 'HIGH']), **gender_label** (str), **education_label** (str), **marriage_label** (str), **age_group** (str), **min_age** (int), **max_age** (int), **min_credit_limit** (Decimal), **max_credit_limit** (Decimal), **sort_by** (Literal['customer_id', 'age', 'c), **sort_order** (Literal['asc', 'desc'])

Response format: JSON response.

Example request

```
GET /api/exports/customers?risk_band=HIGH
Authorization: Bearer <jwt>
```

Illustrative response

```
HTTP 200
Content-Type: text/csv

customer_id,gender_label,age,credit_limit,risk_score,risk_band
1,Female,30,50000,85,HIGH
```

4. Common Status Codes and Notes

200 successful retrieval/update; **201** review created; **204** review deleted; **401** invalid/missing authentication; **403** insufficient role; **404** resource not found; **422** invalid request; **503** database unavailable. Exact errors vary by endpoint.

Customer and review lists support pagination with page >= 1 and page_size 1-100. Risk bands used by customer filters are LOW, MEDIUM, HIGH. Review statuses are Pending, In Progress, Completed; priorities are Low, Medium, High. Deleting reviews requires administrator role.

The project exposes no ETL-control or risk-score calculation HTTP endpoint in the inspected routers. ETL and risk calculations are implemented as backend modules/scripts, not as additional documented APIs.